Step Coupling in the Published pmsimstats Code: The Biomarker Stays Fully Coupled to the Drug Response After the Drug Is Stopped

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Purpose. This note describes a second specific feature of the code that accompanies Hendrickson et al. (2020) [1], the `pmsimstats` package at commit `58b32a9` [2]. In the simulation the biomarker predicts the drug response through a correlation between the two. The code sets that correlation to its full value at every visit where the expected drug response is nonzero. Once carryover is switched on, the expected drug response is nonzero at every off-drug visit after the first exposure, so the biomarker remains as strongly associated with the drug response when the drug has been stopped as when it is being taken. The on-drug versus off-drug contrast then carries no biomarker signal, and the power to detect the biomarker-by-treatment interaction falls to the test size at any positive half-life. The note quotes the code verbatim, derives the size of the effect, works two examples with every number shown, states what the feature does and does not change downstream, and gives a short correction with a check that it reproduces the intended coupling exactly. It is a companion to `docs/53`, which treats the other defect in the same file, the compounding carryover recursion of the drug-response mean. Like `docs/53`, it is written to be read on its own, and in particular by the authors of the article and the original code, who are best placed to confirm or correct the reading given here.

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Notation index and glossary

Notation follows the project's canonical table (`analysis/report/NOTATION.md`) and, for the coupling quantities, the closed-form paper `docs/37`. Symbols marked *local* are defined here (or in `docs/37`) and are not in the canonical table.

Table 1: Notation index for this note

Symbol	Meaning	Status
t	visit index within a randomization path	canonical
D_{it}	binary drug state: 1 on drug, 0 off drug	canonical
t_{od}	cumulative time on drug at a visit (code `tod`); the input to the Gompertz curve	canonical
t_{sd}	time since discontinuation (code `tsd`), cumulative within an off-drug run, zero on drug and before first exposure	canonical
$t_{1/2}$	carryover half-life, in weeks (code `carryover_t1half`)	canonical
λ	decay rate, $\ln 2 / t_{1/2}$ (code `lambda_cor` in pmsimstats-ng)	canonical
B	participant-level biomarker (code `bm`; B_i in `NOTATION.md`); $\bar{B}$ its sample mean	canonical
μ_{bm}, σ_{bm}	biomarker mean and standard deviation, 124.33 and 15.36	canonical (σ_{bm})
BR_t	the biological (drug) response component at visit t	canonical
σ_{BR}	standard deviation of BR_t, 8	canonical
$\mu_{BR,t}$	expected BR at visit t (code `brmeans[p]`)	proposed
μ_{last}	μ_{BR} at the last on-drug visit before an off-drug run	local (`docs/53`)
m, $t_{sd,m}$	position of an off-drug visit within its run, and its cumulative t_{sd}	local (`docs/53`)
c_{bm}	interaction strength of the covariance construct: the on-drug correlation of B with BR_t (code `c.bm`)	canonical
g_t	coupling: the fraction of c_{bm} present at visit t, so that $\text{Cor}(B, BR_t) = c_{bm}\, g_t$	local (`docs/37`)
$\bar{g}_{\text{on}}$, $\bar{g}_{\text{off}}$	mean coupling over a path's on-drug and off-drug post-baseline visits	local (`docs/37`)
$\bar{g}_{\text{on}} - \bar{g}_{\text{off}}$	the coupling gap	local (`docs/37`)
n_{on}, n_{off}	numbers of on-drug and off-drug post-baseline visits of a path	local
$\beta_{bm:D}$	the biomarker-by-treatment interaction: the slope of a participant's on-minus-off outcome contrast on B	canonical
ϕ_t	implied slope of BR_t on B, $c_{bm}\, g_t\, \sigma_{BR}/\sigma_{bm}$ (points of BR per biomarker unit)	local
Y_{it}, BL_i, TV_{it}, PB_{it}	outcome, baseline level and the other two response components	canonical
Δ_i	participant i's on-minus-off outcome contrast, $\bar{Y}_i^{\text{on}} - \bar{Y}_i^{\text{off}} = \sum_t a_t Y_{it}$	local (`docs/37`)
a_t	contrast weights, $1/n_{\text{on}}$ on drug and $-1/n_{\text{off}}$ off drug	local
β_D	treatment main effect (the `Db` coefficient of the published model)	canonical
v	coupling pattern over visits, the vector of g_t	local (`docs/54`)
M	correlation matrix of the response components	local (`docs/54`)
s_f	the scale factor added in 2024 (code `scalefactor`, default 2)	local (`docs/53`)

Symbol	Meaning	Status
c^*_{bm}	ceiling: the largest c_{bm} for which the correlation matrix is positive definite	local (`docs/37`)

Glossary.

- **Coupling.** How strongly the biomarker is tied to the drug response at a visit, measured as the fraction g_t of the on-drug correlation c_{bm} that is present there. The term is ours, not the article's.
- **Step coupling.** The rule of the `58b32a9` code: $g_t = 1$ wherever $\mu_{BR,t} \neq 0$, and $g_t = 0$ elsewhere.
- **Graded coupling.** The rule recommended here: $g_t = 1$ on drug, $g_t = 2^{-t_{sd}/t_{1/2}}$ off drug after first exposure, and $g_t = 0$ before first exposure.
- **Coupling gap.** $\bar{g}_{\text{on}} - \bar{g}_{\text{off}}$, the difference between how strongly the biomarker is tied to the drug response on drug and off drug, averaged over a path's visits.
- **E9, E9b.** The paired-difference statistic of `docs/37`: each participant's mean outcome on drug minus mean outcome off drug, regressed on the biomarker by ordinary least squares. E9b counts the baseline as one more off-drug visit, as the published analysis does. Both have exact closed-form moments, and E9b reproduces the mean estimates and the Hybrid power of the published mixed-model analysis (`docs/37`, Section 6).
- **Hybrid design.** The design that the article calls the "N-of-1 design": eight visits at weeks 4, 8, 9, 10, 11, 12, 16 and 20, with open-label titration, a blinded discontinuation and a brief crossover, and four randomization paths (`pathA` to `pathD`, here paths 1 to 4), with 18, 18, 17 and 17 of $N = 70$ participants.
- **OL+BDC.** Open-label treatment followed by blinded discontinuation; two paths. **CO.** Two-period crossover, eight visits 2.5 weeks apart; two paths, one of which starts on placebo.
- **Ceiling.** The largest c_{bm} at which the correlation matrix the code builds is positive definite; above it the code repairs the matrix silently (`docs/36`).
- **The published process.** The data-generating process of the `58b32a9` code (`NOTATION.md`). **The refined process.** The pmsimstats-ng data-generating process, which among other changes uses graded coupling (`NOTATION.md`). Both are used only where a name is needed.

Evidence labels. Each claim carries one of four labels: *quoted* (text reproduced from a source), *inspected* (read from source code), *derived* (follows algebraically) or *verified* (computed by running code, with the script named in Section 13).

1. Summary for the busy reader

What the code does. In `R/generateData.R` at `58b32a9`, lines 131 to 138 set the correlation between the biomarker and the drug response at visit t to c_{bm} if the expected drug response at that visit is nonzero, and leave it at 0 otherwise (inspected). The expected drug response used in the test already includes carryover. For any $t_{1/2} > 0$ it is positive at every off-drug visit after the first exposure, so those visits receive the full on-drug correlation (derived; verified in all 40 path and half-life cases of Section 9.3).

What it should do. The biomarker is meant to predict the size of the drug effect. A drug effect that has faded cannot be predicted at full strength, so the coupling should fade with it: $\text{Cor}(B, BR_t) = c_{bm}$ on drug, $c_{bm}\, 2^{-t_{sd}/t_{1/2}}$ off drug after exposure, and 0 before first exposure. The article does not state an off-drug rule (Section 3), so this graded rule is our reading of the pharmacological intent.

This is a defect in the coupling, triggered by the mean. The expected drug response itself is not at fault here; its carryover is the subject of `docs/53`. What goes wrong is the second moment: the code uses "is the mean nonzero?" as a switch for the correlation, and carryover turns the switch on everywhere after exposure.

How large. With full coupling on and off drug, the coupling gap is 0 and the slope of the on-minus-off contrast on the biomarker is exactly 0 (derived). In the closed form for the Hybrid design at $N = 70$, $c_{bm} = 0.25$, power is 0.721 without carryover and the test size, 0.050, at every half-life from 0.0125 weeks upward; for E9b, the proxy for the published analysis, 0.654 falls to 0.081 (verified). On Hendrickson's own code with the published analysis, power falls from 0.628 at $t_{1/2} = 0$ to 0.078 at 0.1 week (`docs/53`, Section 8.3; quoted; reproduced in Section 7.3). With graded coupling it is 0.658 at 0.1 week and declines gradually, to 0.486 at 1 week and 0.344 at 2 (quoted, same source). The collapse is a switch, not a decline: it is discontinuous at $t_{1/2} = 0$, and any positive half-life, however small, triggers it, until the half-life is so small that the expected response underflows to exactly zero in floating point (Section 7.2).

What it does not touch. The means, the variances and every correlation not involving the biomarker are unchanged, so any summary that does not use the biomarker, including the population-average drug effect, has the same distribution under either rule (derived; verified, Section 8.3). The rule does change the positive-definiteness ceiling: under step coupling the Hybrid ceiling jumps from 0.256 to 0.893 at any positive half-life, precisely because the coupling no longer varies between visits (verified, Section 8.2).

The fix. Replace the test on the mean with a coupling computed from the drug state and t_{sd}, which `buildtrialdesign()` already supplies:

```
if(d[p]$tod>0){
  g<-1
}else if(d[p]$tsd>0 && modelparam$carryover_t1half>0){
  g<-(1/2)^(d[p]$tsd/modelparam$carryover_t1half)
}else{
  g<-0
}
correlations[n1,'bm']<-modelparam$c.bm*g
correlations['bm',n1]<-modelparam$c.bm*g
```

The patched file reproduces the intended coupling to within 1.1×10^{-16} in all 40 cases checked (verified, Section 9.3). The 2024 upstream revision replaces the step with a ratio of consecutive means, which removes the collapse but fades at half the stated half-life, never couples visit 1, and works only because the means compound (Section 4.5).

2. Scope

This note concerns one statement: the off-drug correlation between the biomarker and the drug response. It does not address the carryover of the drug-response mean (`docs/53`), the compound-symmetry correlation structure, the positive-definiteness repair or the analysis model, except where they interact with the coupling rule; those interactions are stated in Sections 8 and 10.

3. The intended model

The article describes the covariance matrix in one sentence (quoted; [1], Methods, p. 6):

> The covariance matrix was assembled based on a set of modifiable parameters defining the correlation between the baseline biomarker and the BR components, the autocorrelation over time (relating the value of a factor for one participant at one timepoint to the value of that factor for that participant at subsequent timepoints), the correlation at a single time point between the 3 factors, and the variance of each component.

and describes the carryover of the drug response, but not of its correlation with the biomarker (quoted; [1], Methods, p. 6):

> The BR factor was set to zero at times when participants had never been on active drug; however, a carryover effect was built in such that when a participant moved from active drug to being off active drug, the value of the BR at the last timepoint on active drug was exponentially decayed, with the half-life of this decay being maintained as a model parameter.

The parameter is called "the correlation coefficient relating the biomarker to the biologic response to drug" in the caption of Figure 4 ([1], p. 8; quoted), and the package documents `c.bm` as "Correlation between the biomarker and the biologic response" (`R/generateData.R`, line 11; quoted). Neither source says what the correlation should be at an off-drug visit. The rule below is therefore our reading of the intent, not a quotation.

In words: the biomarker predicts how large a participant's drug response is. While the drug is taken, the correlation is c_{bm}. When the drug is stopped, the drug response decays with half-life $t_{1/2}$, and a biomarker that predicts the size of the drug effect should predict a fading effect with fading strength; "the biomarker predicts the size of the drug effect, so it cannot predict a drug effect that is no longer present" (report 01, Methods, "Pharmacological grounding"; quoted). Before the first exposure there is no drug effect and no coupling. Writing the coupling as g_t,

$$\mathrm{Cor}(B, BR_t) = c_{bm}\, g_t, \qquad g_t = \begin{cases} 1, & D_{it} = 1, \\ 2^{-t_{sd,t}/t_{1/2}} = e^{-\lambda t_{sd,t}}, & D_{it} = 0 \text{ after first exposure}, \\ 0, & \text{before first exposure, or } t_{1/2} = 0. \end{cases}$$

The decay of the coupling is the decay the article states for the drug response, so the ratio of the biomarker-related spread of BR_t to its mean stays at its on-drug value off drug (derived; `analysis/report/compendium-mathematics.tex`, section "The pharmacokinetic argument for graded coupling"). The interaction the analysis is meant to detect is the difference between on-drug and off-drug coupling: if the biomarker predicts the response as strongly off drug as on drug, there is no biomarker-by-treatment interaction to find (`docs/37`, Section 2; quoted in substance). Carryover in within-subject designs has a long methodological literature [6]; what is specific here is that carryover must be specified twice, for the mean of the drug response and for its association with the biomarker.

The article's discussion of the result is consistent with this reading: "Although this decrease in power is consistent with expectation, the magnitude of this drop with even short half-life carryover effects underscored the critical nature of this parameter" ([1], pp. 11 to 12; quoted). A decrease in power that grows with the half-life is what graded coupling produces (Section 7); a drop to the test size at 0.1 week is what step coupling produces.

4. The code as published

4.1 The coupling lines

`R/generateData.R` at `58b32a9`, lines 131 to 138 (quoted; permalink):

```
131       # correlation with biomarker
132       for(p in 1:nP){
133         n1<-paste(trialdesign$timeptname[p],"br",sep=".")
134         if(means[which(n1==labels)]!=0){
135           correlations[n1,'bm']<-modelparam$c.bm
136           correlations['bm',n1]<-modelparam$c.bm
137         }
138       }
```

The block sits inside the loop `for(c in cl)` that opens at line 101, so it runs three times with the same effect. The matrix it writes into starts as the identity (lines 98 to 100), so a visit that fails the test keeps a biomarker correlation of exactly 0, and the covariance matrix is formed from it at line 141 (permalink):

```
 97    # Turn this into a matrix of correlations
 98    correlations<-diag(length(labels))
 99    rownames(correlations)<-labels
100    colnames(correlations)<-labels
101    for(c in cl){
...
141    sigma<-outer(sds,sds)*correlations
```

4.2 Where the means in the test come from

The vector `means` tested on line 134 is assembled at lines 73 to 95; its drug-response part is built at lines 83 to 93 (quoted; permalink):

```
 83        brmeans<-modgompertz(d$tod,rp$max,rp$disp,rp$rate)
 84        if(nP>1){
 85          for(p in 2:nP){
 86            if(!d[p]$onDrug){
 87              if(d[p]$tsd>0){
 88                brmeans[p]<-brmeans[p]+brmeans[p-1]*(1/2)^(d$tsd[p]/modelparam$carryover_t1half)
 89              }
 90            }
 91          }
 92        }
 93        means<-c(means,brmeans)
```

with the drug state set at line 55, `d[,onDrug:=(tod>0)]`, and t_{od} and t_{sd} supplied by `buildtrialdesign()` (lines 86 to 106 of that file; quoted and explained in `docs/53`, Section 4.2). Line 88 is the compounding recursion of `docs/53`: it decays the previous visit's already-decayed mean by the cumulative t_{sd}. For the present note only one property of it matters: for $t_{1/2} > 0$ it is strictly positive at every off-drug visit after an exposure, whatever its exact value (derived: the Gompertz mean at the last on-drug visit is positive, and 2^{-x} is positive for every finite x). The same is true of the anchored mean that `docs/53` recommends. The step rule's trigger is therefore pulled by any carryover, compounding or not.

4.3 Line-by-line reading

All statements in this subsection are inspected, except where marked.

- **Line 83.** `brmeans` is the Gompertz mean $G(t_{od})$ at every visit. $G(0) = 0$, so it is 0 at every off-drug visit before the carryover lines run.
- **Lines 85 to 88.** At an off-drug visit after exposure ($t_{sd} > 0$), the mean becomes `brmeans[p-1]` $\times$ $2^{-t_{sd}/t_{1/2}}$. With $t_{1/2} > 0$ this is positive. With $t_{1/2} = 0$, R evaluates $2^{-t_{sd}/0} = 2^{-\infty} = 0$, so the mean stays 0.
- **Line 134.** The test reads the mean of the drug-response variable at visit p (`means[which(n1==labels)]`) and compares it with 0. Without carryover it is nonzero exactly at the on-drug visits, so the test separates on-drug from off-drug visits, as intended. With carryover it is nonzero at every visit from the first exposure onward.
- **Lines 135 and 136.** Every visit that passes the test receives the same correlation, `modelparam$c.bm`; there is no intermediate value. The coupling is two-valued: 0 or 1 in units of c_{bm}.

- **Consequence (derived).** For $t_{1/2} > 0$ the code's coupling is $g_t = 1$ at every on-drug visit and at every off-drug visit after first exposure, and $g_t = 0$ only before first exposure. In the Hybrid and OL+BDC designs every path starts on drug, so after the first visit the coupling is 1 everywhere.

4.4 A coupling defect with a mean trigger

It may be natural to read this as a problem with the means, since the test is written on the means. It is not. The means at the off-drug visits are small and positive, as carryover intends: at $t_{1/2} = 0.1$ week the drug-response mean one week after stopping is 0.0099 points, 0.1% of its last on-drug value of 10.19 (verified, Table 3). The defect is that a quantity of that size switches on a correlation of full strength. The mean decides *whether* the visit is coupled; nothing decides *how much*. Correcting the means, as `docs/53` does, leaves the switch exactly where it was (verified: the pattern of nonzero means is identical under the published and the anchored means in all 40 cases of `docs/53`, Section 8.2).

The coupling enters the data only through the covariance matrix. Under joint normality,

$$E(BR_t \mid B) = \mu_{BR,t} + \phi_t\,(B - \mu_{bm}), \qquad \phi_t = c_{bm}\, g_t\, \frac{\sigma_{BR}}{\sigma_{bm}},$$

so ϕ_t is the number of points by which a participant's expected drug response at visit t rises per unit of biomarker (derived). Under step coupling, at $c_{bm} = 0.25$, $\phi_t = 0.25 \times 8/15.36 = 0.1302$ at every visit after first exposure, whether the drug is on or off, and whether the mean drug response there is 10 points or 0.01 (verified, Table 3). The off-drug mean is the carryover of the drug effect, and it fades; the step rule makes the biomarker's prediction of that effect carry over undiminished.

4.5 What the 2024 revision changed

Three later commits changed `R/generateData.R` (inspected, GitHub commit history and per-commit patches; `docs/53`, Section 4.4). Only one, `8609f12` (2024-05-06, committed by RC Hendrickson with the message "Ron Thomas' edits, currated into working code"; commit), changes the coupling. The current upstream file (`master` at `06dac83`, 2024-11-27), lines 145 to 154 (quoted; permalink):

```
145       # correlation with biomarker
146       for(p in 1:nP){
147         n1<-paste(trialdesign$timeptname[p],"br",sep=".")
148         ## RON THOMAS VERSION:
149         if (p > 1) {
150           n0 <- paste(trialdesign$timeptname[p - 1], "br", sep = ".")
151           mm1 <- means[which(n1 == labels)]
152           mm0 <- means[which(n0 == labels)]
153           correlations["bm", n1] <- correlations[n1, "bm"] <- ifelse(brtest[p], ifelse(brmeans[p] ==
154         }
```

The original lines 134 to 137 follow as comments (lines 156 to 161). `brtest` is set before the carryover lines, at line 87, `brtest <- brmeans == 0`, so it is `TRUE` exactly at the off-drug visits; line 89 names it with `labels[19:26]`, which assumes eight visits (inspected). Reading line 153 (derived, and verified in 40 cases to 1.1×10^{-16}, Section 9.3):

- **On drug, from visit 2:** $g_t = 1$.
- **Off drug, after exposure:** $g_t = \mu_{BR,t}/\mu_{BR,t-1}$, the ratio of consecutive carryover-adjusted means. Under the current recursion with $s_f = 2$ (line 96) that ratio is $2^{-s_f t_{sd,t}/t_{1/2}}$ with the cumulative t_{sd}, so $g_t = 2^{-2t_{sd,t}/t_{1/2}}$: graded coupling at half the stated half-life.

- **Off drug before exposure, or with $t_{1/2} = 0$:** $g_t = 0$.
- **Visit 1:** never assigned, because the block is guarded by `if (p > 1)`; its correlation stays at the identity's 0, even though visit 1 is on drug in every path of the Hybrid and OL+BDC designs. `docs/54`, Section 3.4, analyzes this side effect ("the first occasion is never assigned"; that note calls the 2024 vendored copy `orig` and uses its line numbers).

So the current upstream code no longer collapses: its coupling gap is positive at every half-life (Table 6). It is not the intended rule in three respects (derived; verified, Sections 7 and 10).

1. **It fades twice as fast as the stated half-life**, through the default $s_f = 2$, which `generateSimulatedResults()` does not override (`docs/53`, Section 4.4; inspected).
2. **It never couples visit 1.** The on-drug mean coupling is $(n_{\text{on}} - 1)/n_{\text{on}}$, 0.8 on Hybrid paths 1 and 2 and 0.75 on paths 3 and 4, so the interaction slope without carryover is -0.101 instead of -0.130, and closed-form E9 power is 0.495 instead of 0.721 (verified, Table 6). The direct graded rule assigns c_{bm} at visit 1.
3. **It is right only because the means compound.** The ratio of consecutive means equals $2^{-s_f t_{sd,t}/t_{1/2}}$ only under the compounding recursion of `docs/53`. If the means are corrected to the anchored form and this rule is kept, the ratio becomes a constant factor per visit and the coupling stops fading after the first off-drug visit (Section 10, Table 11; `docs/53`, Section 9.5).

Table 2: Coupling g_t on Hybrid path 1 at $t_{1/2} = 1$ week under the 58b32a9 code, the current upstream code and the intended rule (verified)

Visit	D_{it}	t_{sd}	58b32a9 (step)	2024 upstream (ratio, $s_f = 2$)	Intended (graded)
OL1	1	0	1	0	1
OL2 to BD2	1	0	1	1	1
BD3	0	1	1	0.250	0.500
BD4	0	2	1	0.0625	0.250
COd	1	0	1	1	1
COp	0	4	1	0.0039	0.0625

5. Why it removes the interaction signal

The algebra is short, and it is exact for the paired-difference statistic E9 of `docs/37` (all statements derived; the source is `docs/37`, Sections 4.2 and 5.1).

For participant i, let $\Delta_i = \bar{Y}_i^{\text{on}} - \bar{Y}_i^{\text{off}} = \sum_t a_t Y_{it}$, with weights $a_t = 1/n_{\text{on}}$ on drug and $-1/n_{\text{off}}$ off drug. Since $Y_{it} = \text{BL}_i - (TV_{it} + PB_{it} + BR_{it})$ and only BR_{it} is correlated with the biomarker, with covariance $c_{bm} g_t \sigma_{bm} \sigma_{BR}$,

$$\beta_{bm:D} = \frac{\text{Cov}(\Delta, B)}{\text{Var}(B)} = -\sum_t a_t \, \phi_t = -c_{bm} \frac{\sigma_{BR}}{\sigma_{bm}} \Big(\bar{g}_{\text{on}} - \bar{g}_{\text{off}} \Big).$$

The interaction slope is the implied biomarker slope on drug minus the implied biomarker slope off drug, and it is proportional to the coupling gap. Without carryover $\bar{g}_{\text{on}} = 1$, $\bar{g}_{\text{off}} = 0$, and $\beta_{bm:D} = -c_{bm}\sigma_{BR}/\sigma_{bm} = -0.1302$ at $c_{bm} = 0.25$.

Step coupling. For $t_{1/2} > 0$ every visit after first exposure has $g_t = 1$ (Section 4.3), so in the Hybrid and OL+BDC designs $\bar{g}_{\text{on}} = \bar{g}_{\text{off}} = 1$, the gap is 0, and

$$\beta_{bm:D}^{\text{E9}} = 0 \quad \text{for every } t_{1/2} > 0.$$

The biomarker slope is the same on and off drug, and the on-minus-off contrast removes it. E9b, which counts the baseline (where $g_0 = 0$) as one more off-drug visit, keeps a remnant:

$$\beta^{\mathrm{E9b}}_{bm:D} = -c_{bm}\frac{\sigma_{BR}}{\sigma_{bm}}\left(1 - \frac{n_{\mathrm{off}}}{n_{\mathrm{off}}+1}\,\bar{g}_{\mathrm{off}}\right) = -c_{bm}\frac{\sigma_{BR}}{\sigma_{bm}}\cdot\frac{1}{n_{\mathrm{off}}+1},$$

one quarter of the full slope on Hybrid paths 1 and 2 ($n_{\mathrm{off}} = 3$) and one fifth on paths 3 and 4. That remnant, pooled -0.029, is what the published mixed model estimates under carryover: -0.026 to -0.032 in simulation (`docs/37`, Section 5.1; quoted).

Graded coupling. $\bar{g}_{\mathrm{off}} = n_{\mathrm{off}}^{-1}\sum_{\mathrm{off}} 2^{-t_{sd,t}/t_{1/2}}$ is strictly increasing in $t_{1/2}$ and tends to 0 as $t_{1/2} \to 0$, so the gap starts at 1 and declines smoothly. Because every off-drug visit in Hybrid has $t_{sd} \geq 1$ week, at $t_{1/2} = 0.1$ week $\bar{g}_{\mathrm{off}} \leq 2^{-10} \approx 0.001$, and the gap is still above 0.999. As $t_{1/2} \to \infty$, $\bar{g}_{\mathrm{off}} \to 1$: the step rule is the infinite-half-life limit of graded coupling, applied at every positive half-life (`docs/37`, Section 5.2).

Power depends on the coupling only through the gap. The variance of Δ involves only the response variances and correlations, which neither rule touches, so it does not depend on $t_{1/2}$ or on the coupling rule (`docs/37`, Proposition 1). The residual variance is $\mathrm{Var}(\Delta) - \beta^2_{bm:D}\sigma^2_{bm}$, and power is a strictly increasing function of $|\beta_{bm:D}|$, hence of the gap (`docs/37`, Section 5.4). Neither which off-drug visits are coupled nor how the coupling is spread over them matters for E9; only the average does. Under step coupling the gap is 0 at every positive half-life, the bottom of that curve.

6. Worked example 1: Hybrid path 1

6.1 Set-up

The design is the article's Hybrid design as defined in the package's publication vignette (`vignettes/Produce_Publication_R` lines 48 to 61 at `58b32a9`): path 1 (`pathA`) is on drug at visits 1 to 4 and 7 (weeks 4, 8, 9, 10 and 16) and off at visits 5, 6 and 8 (weeks 11, 12 and 20), one, two and four weeks after a discontinuation (inspected). The parameters are those of the article: $c_{bm} = 0.25$ (below the ceiling, so no matrix is repaired), $\sigma_{BR} = 8$, $\sigma_{bm} = 15.36$, and the shipped `extracted_rp` drug-response curve (inspected).

Every coupling below was read from the correlation matrix that the code itself builds: the published `generateData()` and the patched version of Section 9 were run with the multivariate normal sampler replaced by a function that records the mean vector and covariance matrix it is given, and g_t was computed as $\mathrm{Cor}(B, BR_t)/c_{bm}$ from that matrix (verified). The graded column equals $2^{-t_{sd}/t_{1/2}}$ off drug to within 10^{-16}.

6.2 Step by step at $t_{1/2} = 0.1$ week

- **OL1 to BD2 (weeks 4 to 10), on drug.** The Gompertz means are 4.281, 9.223, 9.793 and 10.187 points. Both rules: $g_t = 1$, $\phi_t = 0.1302$.
- **BD3 (week 11), off drug, $t_{sd} = 1$.** Published mean $10.187 \times 2^{-1/0.1} = 10.187 \times 2^{-10} = 0.00995$. It is not zero, so the step rule sets $g_t = 1$ and $\phi_t = 0.1302$: one biomarker standard deviation (15.36 units) raises the expected drug response by 2.0 points, at a visit where the expected drug response itself is 0.01 points. Graded: $g_t = 2^{-10} = 0.000977$, $\phi_t = 0.000127$.
- **BD4 (week 12), off drug, $t_{sd} = 2$.** Published mean $0.00995 \times 2^{-20} = 9.5 \times 10^{-9}$, still not zero: step $g_t = 1$. Graded: $2^{-20} = 9.5 \times 10^{-7}$.
- **COd (week 16), on drug again.** t_{od} restarts at 4 weeks; both rules $g_t = 1$.
- **COp (week 20), off drug, $t_{sd} = 4$.** Published mean $4.281 \times 2^{-40} = 3.9 \times 10^{-12}$: step $g_t = 1$. Graded: $2^{-40} = 9.1 \times 10^{-13}$.

Table 3: Hybrid path 1 at $t_{1/2} = 0.1$ week: published drug-response mean, the step test, coupling and implied biomarker slope ϕ_t (points of BR per biomarker unit) under the step and graded rules, $c_{bm} = 0.25$ (verified)

Visit	Week	D_{it}	t_{sd}	$\mu_{BR,t}$ (published)	$\mu_{BR,t} \neq 0$	g_t, step	g_t, graded	ϕ_t, step	ϕ_t, graded
OL1	4	1	0	4.281	yes	1	1	0.1302	0.1302
OL2	8	1	0	9.223	yes	1	1	0.1302	0.1302
BD1	9	1	0	9.793	yes	1	1	0.1302	0.1302
BD2	10	1	0	10.187	yes	1	1	0.1302	0.1302
BD3	11	0	1	0.00995	yes	1	9.8×10^{-4}	0.1302	1.3×10^{-4}
BD4	12	0	2	9.5×10^{-9}	yes	1	9.5×10^{-7}	0.1302	1.2×10^{-7}
COd	16	1	0	4.281	yes	1	1	0.1302	0.1302
COp	20	0	4	3.9×10^{-12}	yes	1	9.1×10^{-13}	0.1302	1.2×10^{-13}

6.3 The same path at $t_{1/2} = 1$ week

At one week the off-drug means are large enough to see, 5.094, 1.273 and 0.268 points (the compounding values of `docs/53`; the intended anchored values are 5.094, 2.547 and 0.268). The step rule gives the same coupling as at 0.1 week, $g_t = 1$ everywhere; the graded rule gives 0.5, 0.25 and 0.0625, the fraction of the drug effect that remains.

Table 4: Hybrid path 1 at $t_{1/2} = 1$ week: published drug-response mean, the step test, coupling and implied biomarker slope under the step and graded rules, $c_{bm} = 0.25$ (verified)

Visit	Week	D_{it}	t_{sd}	$\mu_{BR,t}$ (published)	$\mu_{BR,t} \neq 0$	g_t, step	g_t, graded	ϕ_t, step	ϕ_t, graded
OL1	4	1	0	4.281	yes	1	1	0.1302	0.1302
OL2	8	1	0	9.223	yes	1	1	0.1302	0.1302
BD1	9	1	0	9.793	yes	1	1	0.1302	0.1302
BD2	10	1	0	10.187	yes	1	1	0.1302	0.1302
BD3	11	0	1	5.094	yes	1	0.500	0.1302	0.0651
BD4	12	0	2	1.273	yes	1	0.250	0.1302	0.0325
COd	16	1	0	4.281	yes	1	1	0.1302	0.1302
COp	20	0	4	0.268	yes	1	0.0625	0.1302	0.0081

6.4 The contrast slope

Path 1 has $n_{\text{on}} = 5$ and $n_{\text{off}} = 3$. By Section 5, $\beta^{\text{E9}}_{bm:D} = -0.1302\,(\bar{g}_{\text{on}} - \bar{g}_{\text{off}})$:

- **Step, any $t_{1/2} > 0$.** $\bar{g}_{\text{on}} = \bar{g}_{\text{off}} = 1$, gap 0, slope $-0.1302 \times 0 = 0$. E9b: $-0.1302 \times (1 - \frac{3}{4} \times 1) = -0.0325$.
- **Graded, $t_{1/2} = 0.1$.** $\bar{g}_{\text{off}} = (2^{-10} + 2^{-20} + 2^{-40})/3 = 0.000326$, gap 0.99967, slope -0.13015. E9b -0.13016.
- **Graded, $t_{1/2} = 1$.** $\bar{g}_{\text{off}} = (0.5 + 0.25 + 0.0625)/3 = 0.2708$, gap 0.7292, slope $-0.1302 \times 0.7292 = -0.0949$. E9b: $-0.1302 \times (1 - \frac{3}{4} \times 0.2708) = -0.1037$.

The same slopes were computed a second way, directly from the code's covariance matrix as $\mathrm{Cov}(\Delta, B)/\mathrm{Var}(B)$; the two agree to within 10^{-16} (verified). Under the step rule the direct value is -2×10^{-17}, zero to rounding.

Table 5: Hybrid path 1: mean coupling, coupling gap and slope of the on-minus-off contrast on the biomarker, step and graded rules (verified)

Rule	$t_{1/2}$ (weeks)	$\bar{g}_{\mathrm{on}}$	$\bar{g}_{\mathrm{off}}$	Gap	E9 slope	E9b slope
Either	0	1	0	1	−0.1302	−0.1302
Step	0.1	1	1	0	0	−0.0325
Graded	0.1	1	0.00033	0.99967	−0.1302	−0.1302
Step	1	1	1	0	0	−0.0325
Graded	1	1	0.2708	0.7292	−0.0949	−0.1037

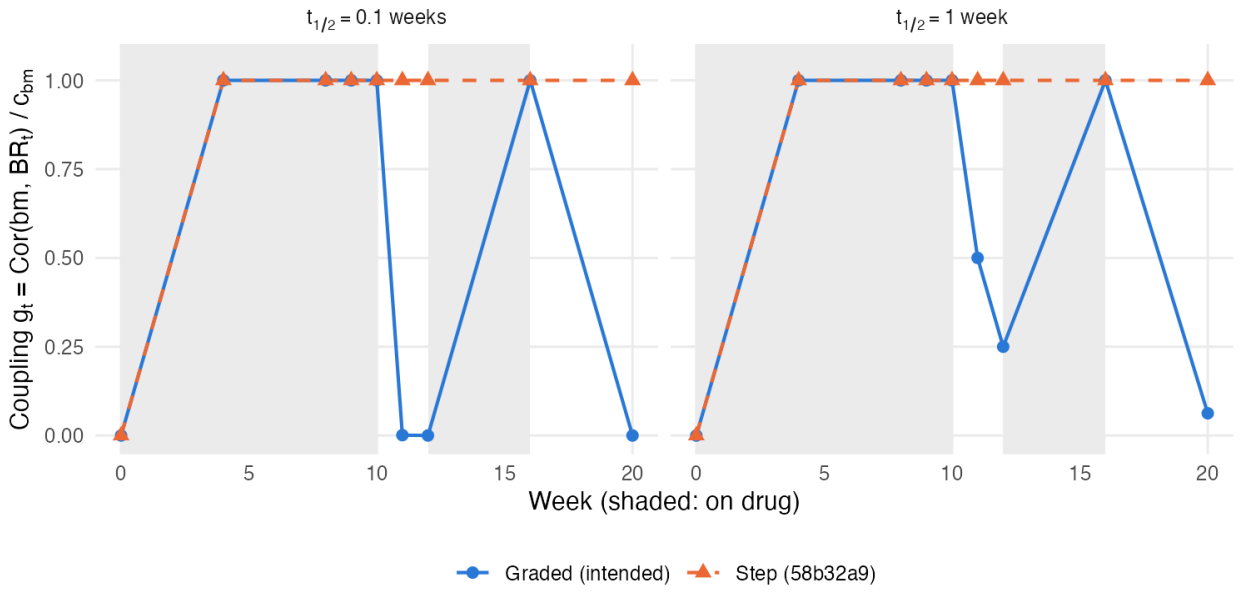


Figure 1: Hybrid path 1: coupling g_t, the fraction of c_{bm} present in the correlation between the biomarker and the drug response, under the published step rule (dashed, triangles) and graded coupling (solid, circles), at $t_{1/2} = 0.1$ week (left) and 1 week (right). Points are visits; baseline is plotted at 0; shaded intervals are on drug. The step rule is 1 at every visit after the first exposure, whatever the half-life, so there is no on-drug against off-drug contrast for the analysis to detect.

7. Worked example 2: power

7.1 Closed form

Table 6 gives closed-form power for the Hybrid design at $N = 70$ and $c_{bm} = 0.25$, computed from the code's own mean vector and covariance matrix for each path and pooled over the four paths with the published allocation (18, 18, 17, 17), following `docs/37`, Section 4.4 (verified). It reproduces the `docs/37` values (its Section 5) to the three decimals printed, from an independent implementation that reads the matrices from Hendrickson's code rather than from a reconstruction.

Table 6: Closed-form power of the paired-difference statistic by coupling rule and $t_{1/2}$, Hybrid design, $N = 70$, $c_{bm} = 0.25$, compound symmetry (verified)

Rule	Statistic	$t_{1/2} = 0$	0.1	0.2	0.5	1	2	4
Step (58b32a9)	E9	0.721	0.050	0.050	0.050	0.050	0.050	0.050
	E9b	0.654	0.081	0.081	0.081	0.081	0.081	0.081
Graded (intended)	E9	0.721	0.721	0.712	0.631	0.473	0.278	0.140
	E9b	0.654	0.653	0.647	0.586	0.473	0.328	0.210
2024 upstream (ratio)	E9	0.495	0.495	0.495	0.475	0.399	0.260	0.127
	E9b	0.438	0.438	0.438	0.424	0.372	0.273	0.166

Two features stand out. Under the step rule power is flat at every positive half-life, at the test size for E9 and at the baseline remnant for E9b; the magnitude of the half-life plays no role. Under graded coupling power is unchanged at the article's half-lives (0.721 at 0.1 week, 0.712 at 0.2) and declines only when carryover is long compared with the one-week spacing of the off-drug visits. The coupling gap, patient-weighted over paths, is 0.9997 at 0.1 week, 0.991 at 0.2, 0.906 at 0.5, 0.754 at 1 and 0.551 at 2 under graded coupling, and 0 at every positive half-life under step coupling (verified; the graded values agree with `docs/37`, Section 5.4).

7.2 The boundary: $t_{1/2} = 0$ and very small half-lives

At $t_{1/2} = 0$ the two rules agree exactly: there is no carryover, the off-drug means are exactly 0 (Section 4.3), and the graded rule's last branch also gives 0 (verified; identical matrices). The step rule's collapse is therefore a discontinuity at $t_{1/2} = 0$: the limit of its power as $t_{1/2} \to 0^+$ is 0.081 (E9b), not the 0.654 it takes at 0 (derived).

At very small half-lives the floating-point arithmetic intervenes. The smallest positive double-precision number is about 4.9×10^{-324}, or 2^{-1074} [5]; a decayed mean smaller than that is stored as exactly 0, and the step test then reads it as "no carryover". The published mean at the m-th off-drug visit of a run is $\mu_{\text{last}}\, 2^{-(t_{sd,1}+\cdots+t_{sd,m})/t_{1/2}}$ (`docs/53`, Section 5), so it underflows when the summed exponent exceeds about 1077 (derived). The half-life below which each visit's mean is exactly 0 was found by bisection on the restated recursion and confirmed on the code itself just above and below each threshold (verified): in the Hybrid design 0.00093 weeks for the first off-drug visit of every run with $t_{sd} = 1$, up to 0.0121 weeks for path 4's fourth off-drug visit; in OL+BDC up to 0.0056; in CO up to 0.0232. Above 0.0121 weeks no Hybrid visit underflows, and the step rule's power is exactly its plateau. Table 7 shows the transition.

Table 7: The boundary of the step rule: step coupling at each path's off-drug visits (1 coupled, 0 not), patient-weighted coupling gap and closed-form E9b power at very small half-lives, Hybrid, $N = 70$, $c_{bm} = 0.25$ (verified)

$t_{1/2}$ (weeks)	Step coupling at the off-drug visits, paths 1 / 2 / 3 / 4	Gap, step	E9b, step	E9b, graded
0	0 0 0 / 0 0 0 / 0 0 0 0 / 0 0 0 0	1	0.654	0.654
0.0005	0 0 0 / 0 0 0 / 0 0 0 0 / 0 0 0 0	1	0.654	0.654
0.001	1 0 0 / 1 0 0 / 1 0 0 0 / 1 0 0 0	0.707	0.437	0.654

$t_{1/2}$ (weeks)	Step coupling at the off-drug visits, paths 1 / 2 / 3 / 4	Gap, step	E9b, step	E9b, graded
0.005	1 1 1 / 1 1 0 / 1 1 0 1 / 1 1 0 0	0.268	0.170	0.654
0.01	1 1 1 / 1 1 1 / 1 1 1 1 / 1 1 1 0	0.061	0.096	0.654
0.0125 and above	1 1 1 / 1 1 1 / 1 1 1 1 / 1 1 1 1	0	0.081	0.654 at 0.0125

At 0.0005 weeks (about 5 minutes) every off-drug mean underflows, and the step rule happens to reproduce the no-carryover result; at 0.001 weeks only the first visit of each run survives; by 0.0125 weeks nothing underflows and the collapse is complete. Graded coupling is unaffected throughout: its off-drug coupling is at most $2^{-1/0.01} = 7.9 \times 10^{-31}$ at 0.01 week, and its own underflow, when $2^{-t_{sd}/t_{1/2}}$ falls below 2^{-1074}, replaces a negligible number by zero (verified). The step rule thus has three regimes in $t_{1/2}$, set by the visit schedule and the limits of double precision rather than by the model: no carryover (exactly 0, or small enough to underflow), a ragged transition below about 0.012 weeks, and the collapse above it. Figure 1 of `docs/37` starts at 0.02 weeks for this reason (`docs/37`, Section 5.1; quoted in substance).

The thresholds depend on the recursion. With the anchored means of `docs/53` the mean at a visit underflows when $t_{sd}/t_{1/2}$ alone exceeds about 1077, so path 4's fourth off-drug visit ($t_{sd} = 7$) would underflow below 0.0065 weeks rather than 0.0121 (derived). This is the only way in which the recursion touches the step rule, and it lies entirely below the half-lives of any practical interest.

7.3 Hendrickson's own code

The closed form is for E9 and E9b, not for the published mixed model. Three simulations ran the published `generateData()` and the published `lme_analysis()` (the `lmer` model of `lme4` [4], with `useDE = FALSE`) on the Hybrid design at $N = 70$, $c_{bm} = 0.25$, without censoring, 500 replicates per cell, with the same seed for each version at a half-life (Table 8).

Table 8: Power of the published analysis for the interaction on data from Hendrickson's generateData() as published (step), with graded coupling only, and in the 2024 upstream version; Hybrid, $N = 70$, $c_{bm} = 0.25$, no censoring, 500 replicates per cell. Run A: `14-graded-patch-58b32a9.R` (verified, stored output). Run B: `docs/53`, Section 8.3 (quoted). Run C: `55-power-check.R` (verified). Monte Carlo standard error at most 0.023

$t_{1/2}$ (weeks)	Published, run A	Graded, run A	Published, run B	Graded, run B	Published, run C	Graded, run C	2024 upstream, run C
0	0.668	0.668	0.628	0.628	0.670	0.670	0.456
0.001					0.464	0.660	0.482
0.01					0.098	0.676	0.458
0.1	0.126	0.638	0.078	0.658	0.100	0.656	0.464
0.2	0.092	0.656	0.082	0.670			
1	0.094	0.462	0.088	0.486	0.076	0.448	0.368
2			0.078	0.344			

Four things follow (verified unless stated).

- **At $t_{1/2} = 0$ the published and graded versions coincide** in every run, as they must: identical matrices and a shared seed give identical data and identical decisions.

- **At any practical half-life the published code collapses and the patch does not.** At 0.1 week published power is 0.078 to 0.126 in the three runs, against 0.638 to 0.658 with graded coupling; at 1 week, 0.076 to 0.094 against 0.448 to 0.486. The spread between runs is within about two combined Monte Carlo standard errors except run A at 0.1 week, which `docs/37` (Appendix A) discusses.
- **The boundary of Section 7.2 appears in the published code itself.** At 0.001 week, where only the first off-drug visit of each run escapes underflow, published power is 0.464, between the no-carryover value and the plateau, as the closed form predicts (E9b 0.437); at 0.01 week it is already 0.098 (closed form 0.096).
- **The 2024 data generator does not collapse, but loses power at every half-life.** Its power is 0.456 to 0.482 at half-lives up to 0.1 week, against 0.670 for the published and graded versions without carryover, because visit 1 is never coupled (Section 4.5); at 1 week it is 0.368 against 0.448 for graded coupling. The mean interaction estimate is -0.102 at $t_{1/2} = 0$, against -0.131 for the other two versions, matching the closed-form -0.101.

No matrix was repaired in any cell of run C.

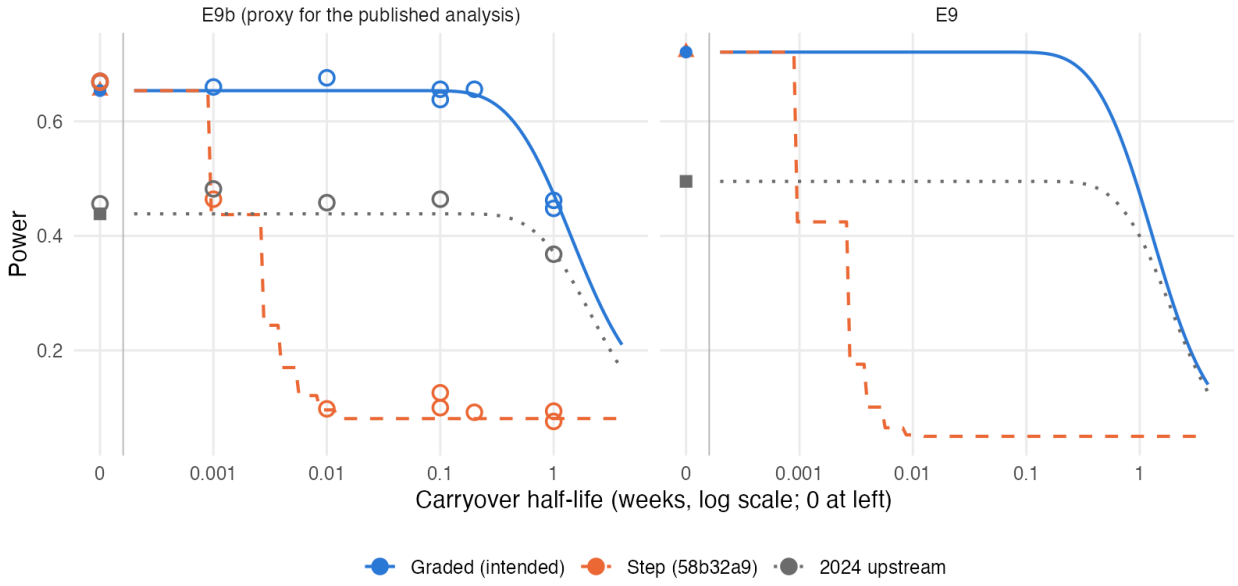


Figure 2: Power for the biomarker-by-treatment interaction against the carryover half-life on a logarithmic axis (the point at the left edge is $t_{1/2} = 0$). Lines: closed form for E9b, the proxy for the published analysis (left), and for E9 (right), under the published step rule, graded coupling and the 2024 upstream ratio rule. Open circles (left panel): power of the published `lmer` analysis on data from Hendrickson's own `generateData()` in each version, runs A and C of Table 8. Hybrid design, $N = 70$, $c_{bm} = 0.25$. The step rule's power is a switch between its value at 0 and its plateau, with a ragged transition below about 0.012 weeks caused by floating-point underflow; graded coupling declines smoothly.

8. Consequences

8.1 What changes: interaction power under carryover

Every carryover scenario of the article's Figure 4B uses $t_{1/2} = 0.1$ or 0.2 week (the `coremodelparams` grid, vignette line 318; inspected). In each of them the step rule leaves no on-drug against off-drug contrast in the Hybrid and OL+BDC designs, and half of it in CO, where the placebo-first path's off-drug visits come before any exposure and keep $g_t = 0$ (`docs/37`, Section 5.7; quoted). This is the pattern the article reports:

"a precipitous decline in power in both the N-of-1 and, to a lesser but still very significant degree, the open label + blinded discontinuation design … A decrease in power in the cross over design was also seen, but this was significantly less severe" ([1], p. 7; quoted). Rerun on the published code at $c_{bm} = 0.25$ for all four designs, both sample sizes and every censoring pattern, graded coupling removes the collapse in each design (report 01, Results, "The correction in the published code", and its Figure 4B replication; quoted).

At the article's own $c_{bm} = 0.6$ there is a further asymmetry. In the Hybrid design the correlation matrix is not positive definite at $t_{1/2} = 0$ on any path, and it is positive definite at $t_{1/2} = 0.1$ on every path (verified). The code repairs the matrix silently when it is not (lines 145 to 150), so the no-carryover column of Figure 4B was simulated from repaired matrices, whose on-drug correlation is 0.54 to 0.57 rather than 0.6 (`docs/36`, Section 4.3; quoted), and the carryover columns from unrepaired matrices with full coupling at every visit (derived from Section 8.2). The comparison across those columns is therefore a comparison between two different models, neither of which is the stated one.

8.2 What changes: the positive-definiteness ceiling

A single biomarker can be correlated with a set of partly independent visits only up to a limit, the ceiling $c^*_{bm} = (v^\top M^{-1} v)^{-1/2}$, with v the coupling pattern and M the correlation matrix of the response components (`docs/36`, Section 4; derived there). The coupling rule changes v, so it changes the ceiling (Table 9; verified, with the code's own matrices).

Table 9: Positive-definiteness ceiling of c_{bm} (minimum over paths) by design, coupling rule and $t_{1/2}$, compound symmetry, $\rho = 0.8$ (verified)

Design	Rule	$t_{1/2} = 0$	0.1	0.2	0.5	1	2
Hybrid	Step	0.256	0.893	0.893	0.893	0.893	0.893
	Graded	0.256	0.256	0.257	0.274	0.307	0.369
	2024 upstream	0.256	0.256	0.256	0.260	0.273	0.295
OL+BDC	Step	0.262	0.893	0.893	0.893	0.893	0.893
	Graded	0.262	0.262	0.265	0.289	0.348	0.469
	2024 upstream	0.256	0.256	0.256	0.260	0.274	0.307
CO	All three	0.256	0.256	0.256	0.256	0.256	0.256

Under the step rule the ceiling jumps from about 0.26 to 0.893 at any positive half-life, because a coupling that is constant over visits aligns with the large person-level component of the compound-symmetric structure and is cheap to accommodate (`docs/36`, Section 4.2; `analysis/report/compendium-mathematics.tex`, Table "Design-level ceilings by configuration"; both quoted, and reproduced here). The cells that stop needing repair are exactly the cells that have lost their signal. Under graded coupling the ceiling rises only slowly with the half-life; under compound symmetry it stays below 0.3 up to $t_{1/2} = 0.5$, so the article's 0.3 and 0.6 remain infeasible there under any coupling rule. That is the separate positive-definiteness problem of `docs/36` (problem 1), which graded coupling does not solve and is not meant to. In CO the binding path is the placebo-first one, whose pre-exposure visits have zero coupling under every rule, so its ceiling does not move (`docs/34`, Section 4.6; quoted). The 2024 rule's ceilings at $\rho = 0.7$, 0.346, 0.351 and 0.368 in Hybrid at $t_{1/2} = 0$, 0.5 and 1, reproduce those of `docs/54` and the compendium exactly (verified).

8.3 What does not change

- **The means and the response block.** The coupling rule writes only the biomarker row and column of the correlation matrix (lines 135 and 136). The mean vector, the variances and every correlation among the baseline and the three response components are identical under the step and graded rules (inspected; verified, largest difference 0 in all 40 cases). Below the ceiling no repair intervenes, so the joint distribution of the outcomes Y_{it}, marginal over the biomarker, is the same under both rules (derived).

- **The population-average drug effect.** It follows that any summary of the outcomes that does not use the biomarker has the same distribution under both rules: the expected on-minus-off contrast $E(\Delta)$ is identical in every path (verified), and so is the drug effect averaged over participants (derived). In the published mixed model this is the `Db` effect evaluated at the mean biomarker, $\hat{\beta}_D + \hat{\beta}_{bm:D}\bar{B}$; its mean over replicates was -8.24 under all three versions at $t_{1/2} = 0$ and -8.16 and -8.14 under the graded and published versions at 0.1 week, a difference of 0.02 against a Monte Carlo standard error of about 0.04 (Table 10; verified). The interaction itself carries none of the main drug effect: the coupling moves where the biomarker places each participant around the mean response, not the mean response.
- **One qualification.** The published model enters the biomarker uncentered (`Sx~bm+Db+t+bm*Db`; lines 117 and 137 of `R/lme_analysis.R`, inspected), so its `Db` coefficient is the drug effect at a biomarker value of 0, far outside the data (mean 124.33). That coefficient equals the average effect minus $\beta_{bm:D}\,\mu_{bm}$, and so it moves with the interaction: at 0.1 week the mean interaction estimates differ by $-0.128 - (-0.031) = -0.097$, so the `Db` coefficient should differ by about $0.097 \times 124.3 = 12$ points (derived); the simulated means are $+7.73$ (graded) and -4.26 (published), a difference of 12.0 (verified, Table 10). Results that report the `Db` coefficient itself, rather than the effect at the biomarker mean, would be affected.

Table 10: Drug coefficients of the published analysis by data-generator version, run C: the uncentered `Db` coefficient moves with the interaction, the drug effect at the mean biomarker does not (500 replicates per cell; verified)

$t_{1/2}$ (weeks)	Version	Mean `bm:Db` estimate	Mean `Db` coefficient (at $B = 0$)	Mean drug effect at mean biomarker	Its SD over replicates
0	Published	−0.131	8.10	−8.23	0.86
0	Graded	−0.131	8.10	−8.23	0.86
0	2024 upstream	−0.102	4.48	−8.23	0.88
0.1	Published	−0.031	−4.26	−8.14	0.85
0.1	Graded	−0.128	7.73	−8.16	0.86
0.1	2024 upstream	−0.099	4.21	−8.15	0.86
1	Published	−0.032	−2.81	−6.81	0.89
1	Graded	−0.105	6.30	−6.72	0.91
1	2024 upstream	−0.091	3.71	−7.63	0.92

The 2024 version's drug effect at 1 week (-7.63 against -6.81 and -6.72) differs for a different reason: its mean recursion uses $s_f = 2$ (`docs/53`, Section 4.4), which changes the off-drug means, not the coupling (inferred from the code; not separated here).

8.4 Where the step rule could matter more

The following are inferred, not tested here.

- **Analyses with a decayed drug indicator.** An analysis that weights off-drug visits by $2^{-t_{sd}/t_{1/2}}$ (the continuous indicator $D_{bc,it}$ that the article's discussion anticipates, "analysis methods that incorporate an expectation of carryover", [1], p. 12) is built for graded coupling. Under step coupling the true interaction off drug is at full strength, so such an analysis would be misspecified in the opposite direction from the one it guards against.
- **Designs with long off-drug runs.** Under graded coupling a design gains contrast from off-drug visits late in a run, where the coupling has faded. Under step coupling those visits add no contrast at all, so design comparisons made with the published code understate the value of washout periods.
- **Estimating the half-life from the biomarker channel.** Under step coupling the correlation structure carries no information about $t_{1/2}$ except whether it is zero.

9. The fix

9.1 The corrected code

The replacement couples in proportion to the drug effect that remains, in the style of the surrounding code (report 01, Appendix B.1; quoted):

```
  # correlation with biomarker: full on drug, decaying with the drug
  # effect off drug (graded coupling), zero before first exposure
  for(p in 1:nP){
    n1<-paste(trialdesign$timeptname[p],"br",sep=".")
    if(d[p]$tod>0){
      g<-1
    }else if(d[p]$tsd>0 && modelparam$carryover_t1half>0){
      g<-(1/2)^(d[p]$tsd/modelparam$carryover_t1half)
    }else{
      g<-0
    }
    correlations[n1,'bm']<-modelparam$c.bm*g
    correlations['bm',n1]<-modelparam$c.bm*g
  }
```

As a minimal diff against `58b32a9`, which replaces lines 134 to 137 and nothing else:

```
--- a/R/generateData.R
+++ b/R/generateData.R
@@ -131,9 +131,14 @@
     # correlation with biomarker
     for(p in 1:nP){
       n1<-paste(trialdesign$timeptname[p],"br",sep=".")
-      if(means[which(n1==labels)]!=0){
-        correlations[n1,'bm']<-modelparam$c.bm
-        correlations['bm',n1]<-modelparam$c.bm
-      }
+      if(d[p]$tod>0){
+        g<-1
+      }else if(d[p]$tsd>0 && modelparam$carryover_t1half>0){
+        g<-(1/2)^(d[p]$tsd/modelparam$carryover_t1half)
+      }else{
+        g<-0
+      }
+      correlations[n1,'bm']<-modelparam$c.bm*g
+      correlations['bm',n1]<-modelparam$c.bm*g
     }
   }
```

Three details of the replacement are deliberate.

- **The coupling is computed from the trial design, not from the means.** `d` is the trial-design table of line 53, which already carries `tod` and `tsd` from `buildtrialdesign()` (`docs/53`, Section 4.2). No new input is needed, and the rule is independent of how the means are computed.

- **The pre-exposure branch.** `tsd` is 0 both on drug and before first exposure (`buildtrialdesign.R`, lines 104 to 105). The `tod>0` test comes first, so on-drug visits get 1, and the remaining `tsd = 0` visits, the baseline and the placebo-first crossover visits, get 0. A rule written as "coupling equals $2^{-t_{sd}/t_{1/2}}$" would give them $2^0 = 1$, full coupling with no drug ever taken (report 01, Appendix B.4, describes the same defect in an analysis indicator).
- **The $t_{1/2} = 0$ branch.** With no carryover, off-drug visits get 0, so the corrected code reproduces the published matrices exactly at $t_{1/2} = 0$ (verified, Section 7.2).

9.2 The package's version

pmsimstats-ng (version 0.3.0) implements the same rule in `buildSigma()`, `R/generateData.R`, lines 341 to 355 (inspected):

```
if(cc=="br" && dgp_architecture %in% c("mvn", "combined")){
  c_bm_cov<-if(dgp_architecture == "combined") modelparam$c.bm_b else modelparam$c.bm
  for(p in 1:nP){
    n1<-paste(trialdesign$timeptnames[p],"br",sep=".")
    if(d[p]$onDrug){
      correlations[n1,'bm']<-c_bm_cov
      correlations['bm',n1]<-c_bm_cov
    } else if(d[p]$tsd>0 && lambda_cor>0){
      decay<-exp(-lambda_cor * d$tsd[p])
      correlations[n1,'bm']<-c_bm_cov * decay
      correlations['bm',n1]<-c_bm_cov * decay
    }
  }
}
```

with `lambda_cor` $= \ln 2/t_{1/2}$ by default (lines 224 to 229), so `exp(-lambda_cor * tsd)` $= 2^{-t_{sd}/t_{1/2}}$, and pre-exposure visits left at the identity's 0. The rest of that package's process differs from `58b32a9` (it uses AR(1) correlation, among other changes), so it is not a clean test of the coupling change alone; the patched `58b32a9` file is.

9.3 Check

The patched `58b32a9` file was run on every path of the Hybrid, OL+BDC and CO designs (eight paths) at $t_{1/2} \in \{0.1, 0.2, 0.5, 1, 2\}$, 40 cases, and the coupling read from its correlation matrix was compared with the intended rule of Section 3 computed directly. The largest absolute difference was 1.1×10^{-16} (verified). In the same 40 cases the package's `buildSigma()` matched the intended rule to 1.1×10^{-16} (verified), the published file's coupling was exactly the indicator of a nonzero mean (verified), and the published off-drug coupling after exposure was 1 in every case (verified). The mean vectors and response blocks of the published and patched files were identical (Section 8.3). The 2024 upstream file's coupling matched graded coupling at half-life $t_{1/2}/2$ with $g_1 = 0$, the reading of Section 4.5, to 1.1×10^{-16} in the same 40 cases (verified).

On the published analysis the patch restores power at the article's half-lives to within Monte Carlo error of the no-carryover value (Table 8): 0.638, 0.658 and 0.656 at 0.1 week against 0.668, 0.628 and 0.670 without carryover, in three independently seeded runs.

9.4 Why the coupling should not be computed from the means

The step rule and the 2024 ratio rule both read the coupling off the drug-response means. That couples two parts of the model that should be specified separately. The step rule turns a small positive mean into full

coupling. The ratio rule works only while the means compound (Section 10), and needs a previous visit, which is why it never couples visit 1. Computing g_t from the drug state and t_{sd}, as in Section 9.1, keeps the coupling correct whatever the mean model, and leaves the decay family to be varied in one place if desired, as the article anticipates ("the simulations could be easily adapted to incorporate an alternative model", [1], p. 12; quoted).

10. How the two defects interact

The `58b32a9` file has two defects in the carryover: the compounding recursion of the mean (`docs/53`) and the step coupling (this note). They touch disjoint lines (83 to 92 and 134 to 137), and their effects separate cleanly in the published process.

- **The step coupling drives the power collapse.** In the paired simulation of `docs/53`, Section 8.3, correcting the mean alone left power at 0.078 to 0.088 at every positive half-life, and changed no test decision at 0.1 or 0.2 week; replacing the coupling raised it to 0.658 at 0.1 week, with or without the mean correction (quoted).
- **The recursion matters at longer half-lives,** and only for the means: under graded coupling, adding the mean correction changed power by at most 0.002 at 1 and 2 weeks (0.486 against 0.484 and 0.344 against 0.344; quoted, same source).
- **The fixes are independent.** The step rule's trigger depends only on whether the means are nonzero, which is the same under either mean (above the underflow thresholds of Section 7.2), and the graded rule does not read the means at all. The two patches can be applied in either order and give byte-identical files (verified), and the `docs/53` 2x2 simulation shows the corresponding power results (quoted above).

The 2024 upstream code is different, because there the coupling is computed from the means. Its ratio rule $g_t = \mu_{BR,t}/\mu_{BR,t-1}$ reproduces a fading coupling only because the means compound; with the anchored means of `docs/53` the ratio becomes $2^{-s_f(t_{sd,t}-t_{sd,t-1})/t_{1/2}}$, a constant factor per visit (derived; `docs/53`, Section 9.5, gives the derivation and its own check). Table 11 confirms it on the upstream file itself, with the anchored mean substituted and nothing else changed.

Table 11: Off-drug coupling g_t of the 2024 upstream code at $t_{1/2} = 1$ week, as is and with only the means corrected to the anchored form, against the intended graded rule (verified)

Design, path	Off-drug visit (t_{sd})	Upstream as is ($s_f = 2$)	Upstream, anchored means ($s_f = 2$)	Intended, stated $t_{1/2}$
Hybrid, 1	BD3 (1)	0.250	0.250	0.500
	BD4 (2)	0.0625	0.250	0.250
	COp (4)	0.0039	0.0039	0.0625
CO, 1	COb1 (2.5)	0.0313	0.0313	0.177
	COb2 (5)	0.00098	0.0313	0.0313
	COb3 (7.5)	3.1×10^{-5}	0.0313	0.0055
	COb4 (10)	9.5×10^{-7}	0.0313	0.00098

With the means corrected and the ratio rule kept, the crossover's coupling would sit at 0.031 for all four off-drug visits, ten weeks after stopping as one week after, while the intended coupling falls to 0.001. The same holds at $s_f = 1$, where the plateau is 0.177 (verified; `docs/53`, Table 8). So for the current upstream code the two fixes must go together: correcting the recursion requires replacing the ratio rule by the direct rule of Section 9.1, which also restores the coupling at visit 1. With the direct rule in place, the two fixes are again independent.

11. How to verify independently

The snippet below needs only `data.table`, `MASS` and `corpcor`, the packages the original code already uses, and a checkout of `rchendrickson/pmsimstats` at `58b32a9` as the working directory. It does not use pmsimstats-ng. It runs the original `generateData()` with `empirical = TRUE`, for which `MASS::mvrnorm()` returns a sample whose means and covariances equal the supplied ones exactly [3], so the sample correlations are the code's own coupling and the regression slope is the population slope.

```
## Run in a checkout of github.com/rchendrickson/pmsimstats at 58b32a9.
library(data.table); library(MASS); library(corpcor)
for (f in c('utilities.R', 'buildtrialdesign.R', 'generateData.R'))
  source(file.path('R', f))
load('data/extracted_rp.rda'); load('data/extracted_bp.rda')
td <- buildtrialdesign(
  timepoints = c(4, 8, 9, 10, 11, 12, 16, 20),
  timeptnames = c('OL1','OL2','BD1','BD2','BD3','BD4','COd','COp'),
  expectancies = c(1, 1, .5, .5, .5, .5, .5, .5),
  ondrug = list(pathA = c(1, 1, 1, 1, 0, 0, 1, 0)))$trialpaths[[1]]
t_half <- 1; c_bm <- 0.25
mp <- data.frame(N = 200, c.bm = c_bm, carryover_t1half = t_half,
  c.tv = .8, c.pb = .8, c.br = .8, c.cf1t = .2, c.cfct = .1)
## empirical = TRUE: sample means and covariances equal the code's own
dat <- generateData(mp, extracted_rp, extracted_bp, td,
  empirical = TRUE, makePositiveDefinite = TRUE)
br <- paste0(td$timeptnames, '.br')
code_g <- sapply(br, function(v) cor(dat$bm, dat[[v]])) / c_bm
on <- td$tod > 0
graded_g <- ifelse(on, 1, ifelse(td$tsd > 0, 2^(-td$tsd / t_half), 0))
print(data.frame(visit = td$timeptnames, on = on, tsd = td$tsd,
  code = round(code_g, 4), graded = round(graded_g, 4)), row.names = FALSE)
## E9: on-drug mean minus off-drug mean of the outcome, regressed on bm
y <- as.matrix(dat[, td$timeptnames, with = FALSE])
delta <- rowMeans(y[, on]) - rowMeans(y[, !on])
k <- c_bm * extracted_rp[cat == 'br']$sd / extracted_bp[cat == 'bm']$sd
cat(sprintf('E9 slope, code: %.4f; graded rule: %.4f\n',
  coef(lm(delta ~ dat$bm))[2], -k * (1 - mean(graded_g[!on]))))
```

Its output, run against the `58b32a9` files with the `extracted_rp.rda` and `extracted_bp.rda` fetched from GitHub at that commit (verified):

```
 visit    on tsd code graded
   OL1  TRUE   0    1 1.0000
   OL2  TRUE   0    1 1.0000
   BD1  TRUE   0    1 1.0000
   BD2  TRUE   0    1 1.0000
   BD3 FALSE   1    1 0.5000
   BD4 FALSE   2    1 0.2500
   COd  TRUE   0    1 1.0000
   COp FALSE   4    1 0.0625
E9 slope, code: 0.0000; graded rule: -0.0949
```

The `code` column is the step rule: full coupling at the three off-drug visits. Changing `t_half` to 0.1 leaves it unchanged; setting it to 0 makes the off-drug entries 0 and the code's slope −0.1302. Applying the change of Section 9.1 before sourcing makes the `code` column equal the `graded` column and the code's slope equal −0.0949.

12. Limitations

- **One defect, one component.** The note treats the off-drug coupling of the biomarker to the drug response. The carryover of the mean is `docs/53`; the compound-symmetry ceiling, which prevents the article's $c_{bm} = 0.3$ and 0.6 from being simulated as stated under either coupling rule, is `docs/36`.
- **The intended rule is our reading.** The article does not state an off-drug coupling (Section 3). Graded coupling is the reading that matches its description of the carryover and the pharmacology of a predictive biomarker; the authors may hold a different intent, which they are best placed to state.
- **Closed form and simulation scope.** The closed form is exact for E9 and E9b, which reproduce the published analysis's mean estimates and its Hybrid power (`docs/37`, Section 6) but are not that analysis. The simulations cover one design (Hybrid), $N = 70$, $c_{bm} = 0.25$, no censoring and 500 replicates per cell; report 01 extends the published-code comparison to all four designs, both sample sizes and the censoring patterns at 250 replicates.
- **The 2024 code** was examined for its data-generating process only, with the published analysis. Its revised `lme_analysis()` (commit `325314f` and later) was not run here.
- **Section 8.4** is inferred, not tested.

13. Reproducibility

Every number labeled verified in Sections 1 and 4 to 11 was computed by two scripts written for this note, in `analysis/scripts/step-coupling/` and run from the repository root:

- `55-coupling-examples.R`: Tables 2 to 7, 9 and 11, both figures, the 40-case checks of the graded patch, of the package's `buildSigma()` and of the 2024 rule, the underflow thresholds, the ceilings, the marginal-distribution check and the order-of-patches check. It runs each version of `generateData()` with the sampler replaced by a recorder of the mean vector and covariance matrix, so that every value is the code's own. Run time about 7 minutes (421 s elapsed, one process).
- `55-power-check.R`: run C of Table 8 and Table 10 (the drug effect), the published `lme_analysis()` on data from the published, graded and 2024 `generateData()`, 500 replicates per cell. Run time 11.2 minutes on 8 cores.

For example,

```
Rscript analysis/scripts/step-coupling/55-power-check.R --reps 500 --cores 8
Rscript analysis/scripts/step-coupling/55-coupling-examples.R out55
```

The first writes to `analysis/data/quick-sim/step-coupling/`; the second writes the tables to `out55/`, the figures to `docs/figures/`, and reads the stored output of the first and of `analysis/scripts/quick-sim/hendrickson-probl` (run A, `analysis/data/quick-sim/hendrickson-problems/graded-patch/graded-patch-reps500.csv`) for Figure 2. Run B is quoted from `docs/53`, whose scripts are in `analysis/scripts/carryover-recursion/`. The scripts read the `58b32a9` sources from `analysis/data/quick-sim/hendrickson-58b32a9/R/`, which were checked to be byte-identical to the file fetched from GitHub at `58b32a9` (`generateData.R`, with `cmp`), and the 2024 file from `analysis/scripts/quick-sim/hendrickson-original-comparison/vendored-hendrickson-genera` which differs from upstream `06dac83` in one semantically identical line (a modern `data.table` column extraction; checked with `diff`). The parameter files `data/extracted_rp.rda` and `data/extracted_bp.rda` in this repository are identical to those at `58b32a9` (`docs/53`, Section 12).

14. References

1. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: statistical methods and theoretical findings. *Frontiers in Digital Health*. 2020;2:13. doi:10.3389/fdgth.2020.00013.

2. Hendrickson RC. pmsimstats: R package for simulation of aggregated N-of-1 trials. GitHub repository rchendrickson/pmsimstats, commit 58b32a92b99a8beb145de00c7bd8a0880f06f0f0 (2020-11-19); commit 8609f124eef1770763ecbb4a8978a474c382f615 (2024-05-06); current `master` 06dac83f31ca85c994a6a2c1a0b98bc18ef233bd (2024-11-27). https://github.com/rchendrickson/pmsimstats.
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5. Goldberg D. What every computer scientist should know about floating-point arithmetic. *ACM Computing Surveys*. 1991;23(1):5-48. doi:10.1145/103162.103163.
6. Senn S. *Cross-over Trials in Clinical Research*. 2nd ed. Chichester: Wiley; 2002. doi:10.1002/0470854596.